

Double Empathy as a Lens to Understand the Design Space for Inclusive Social Play Between Autistic and Neurotypical Children

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ABSTRACT

Social play (SP) is a play activity that involves social interaction between children. Research has shown mixed groups (MG)¹, which include autistic and neurotypical children, benefit more from mutual social interactions. However, inclusive SP in MG reveals challenges in terms of the differences between groups, by which autism is typically characterized. Previous research addressing these issues either only includes one group or only concerns the barriers for autistic children. Addressing this, we aim to understand the design space to support inclusive SP for MG. For this, we utilize the 'double empathy problem' that aims to focus on interpersonal issues rather than problems of one group. In this preliminary work, we conducted interviews with 6 professionals who support MG during SP and examined barriers and design insights. We argue our findings could assist in developing the design space and methods that aim to promote mutual understanding in SP research for MG.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI); Accessibility.**

KEYWORDS

Social Play, Autism, Mixed Groups, Double Empathy Problem

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¹In this paper, the term mixed group refers to the inclusion of both autistic and neurotypical children.

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1 INTRODUCTION

Social play is an intrinsically motivated and voluntary play activity that involves any level of social interaction[13, 24]. It is a crucial part of children's² development and can help them to learn vital social skills that are important for success in the outside world[1, 8]. Unsuccessful or low-quality social interaction during play can frequently lead to feelings of loneliness and can negatively impact mental health[19, 27, 30]. However, children with certain differences, such as autistic children³ who would benefit from experiencing social play with their peers and developing reciprocal friendships, face barriers to be included in social play due to their differences (i.e. differences in communication styles) as well as other environmental, attitudinal, and institutional barriers[20]. Previous research has shown that children interacting within mixed groups of autistic and neurotypical⁴ individuals benefit more from interactions that occur in these settings (i.e. more complex coordinated play and cohesion) compared to non-mixed groups[4]. Yet, these studies also present some barriers to participation in social play that some children - and particularly autistic children - may face (i.e. meltdowns or fallout)[15]. Furthermore, the design knowledge to address these barriers or assist researchers in creating relevant technologies is scarce[13]. Therefore, there is a need for enabling and supporting more social play research and technologies that can support inclusive play for mixed groups. We recognize that the notion of 'inclusion' within education is a complex field of study and understand it to mean that children should have opportunities for equal participation in educational and play activities[16]. Practices that support inclusion can include breaking down barriers to this participation, as well as 'collaborative negotiation, flexibility, and creativity'[25]. With this in mind, our overall goal is to find ways to support and facilitate inclusive social play for mixed groups of autistic and neurotypical children.

²We explicitly refer to children that are of primary school age between 4-12, according to the age group of our participants' pupils (see Methodology).

³

We choose to use label-first language (autistic children) over person-first (people with autism) because it is preferred by self-advocates[18].

⁴We intentionally use the term neurotypical, which targets individuals with typically developing brains. The term non-autistic may include some of the neurodivergent population (people with ADHD or dyslexia) who are not explicitly represented in this work.

However, existing research on interactions between mixed groups tends to situate barriers to social play with the characteristics of autism. These studies often view autism through the medical model of disability, which views differences characterized by autism as deficits located within the individuals that require intervention to align with more typical development[2, 11]. Many of these studies place the responsibility of adjustment on the autistic child, often through intervention approaches that aim to teach and adjust social skills[5]. There are fewer studies that emphasize the self-driven aspect of social play or the diversity in mixed groups[13, 28, 29]. These researchers exerted effort to understand autistic children and create facilitators for social play involving them in the design process via participatory design methods[13]. However, these more recent approaches often only include groups of autistic or neurodivergent children[13]. In addition, the findings of these studies appear to only concern designing for or solving the problems of autistic or neurodivergent children without necessarily considering all children in a mixed group setting[13, 28].

Presenting findings that only relate to or consider the differences of autistic children emerges as a common theme in mixed group research[13, 28]. One way to possibly approach and address this shortcoming is using Milton's concept of the 'double empathy problem'[21]. The double empathy problem suggests problems that might occur from social interactions between different groups of people do not reside in any one group (i.e. the differences of autistic children) but are based on interpersonal issues, which affect both groups mutually (i.e. mutual understanding in mixed groups)[22]. We propose these differences, challenges, or problems could be considered and reflected across all children's experiences and perspectives to help examine mutual interpersonal issues. We are interested in exploring the extent to which the insight from the double empathy problem can help researchers, and in particular, our work, to allow the mutual perspectives of mixed groups in inclusive social play research to emerge.

In this paper, we present preliminary work to reach our goal of designing technologies to support and facilitate inclusive social play for mixed groups of autistic and neurotypical children. Our aim is to understand the design space for inclusive social play for mixed groups. To this end, we conducted semi-structured interviews with 6 professionals who work with both autistic and neurotypical children (i.e. teaching assistants and special educational needs coordinators) on their experience of facilitating inclusive social play to identify the barriers and highlight the opportunities for autistic and neurotypical children in mixed groups. We present 4 themes and 2 initial design insights that could assist in enabling future social play research and technologies that provide opportunities for inclusive participation of all children and developing a design space and methods that aim to promote mutual understanding in mixed groups.

2 BACKGROUND & PREVIOUS WORK

2.1 Social Play & Autism

Social play involves interacting with one or more peers during any form of play, such as constructive, physical, or pretend[17]. Most forms of social play involve a certain level of communication between participants, which can be through play interactions

themselves or verbal and visual mediums[24]. Social play has been shown to have numerous positive effects on development that follow a child throughout their lives[6]. Friendships can emerge from successful social play, which can help a child avoid feelings of loneliness and improve their self-confidence[3].

Autism is a complex and contested phenomenon that is diagnosed by differences in sensory processing, interaction, and communication. Autism is also nuanced and presents differently across individuals[12]. These differences can challenge the development of social skills for young autistic individuals and also affect mutual understanding with their neurotypical peers[15]. Research suggests autistic children's play patterns can lead them to play alone and isolate themselves from the group, resulting in less opportunity to develop important social skills[32]. Regardless, research has also shown that autistic children do possess the desire to socialize, share experiences and develop genuine friendships, especially through play[23, 32]. Both autistic and neurotypical children's perceptions of and desires for friendship are crucial to consider to ensure a mutually satisfying social play environment[9]. Previous research demonstrated, mixed groups, which include both autistic and neurotypical children, have a significant positive effect on both groups' play patterns and social interactions in terms of responsiveness, social orientation and cohesion, and language skills over non-mixed groups[4]. Therefore, it is essential to enable and support social play for mixed groups.

2.2 Social Play for Mixed Groups of Autistic and Neurotypical Children

Some schools may offer social skill support sessions to teach autistic children how to understand and mimic neurotypical social play behavior[7]. This approach places the burden on deciphering neurotypical social situations primarily on the autistic child, rather than focusing on developing mutual understanding among children and professionals[7]. One such example is Auti, an assistive robotic toy designed to encourage favorable social play interactions in autistic children through positive reinforcement and discourage undesirable behavior by removing those rewards[2]. Although Auti was successful in encouraging positive play behaviors, this technology was designed to employ a controversial therapy designed to 'correct' undesired behaviors in autistic children[26]. In addition, this also requires a degree of support from adults which does not follow natural play solely driven by the choices of the children involved.

Only a few studies on designing social play technologies do not aim to change the 'undesired behaviors' of the participating autistic children. Frauenberger et al. employed participatory design (PD) to create social play technologies for mixed groups that did not involve any intervention-based approaches[13]. Several design insights were described to aid in PD with mixed groups, such as balancing the openness and structure of the PD sessions, offering space for solitary time to emotionally regulate, and ensuring the amount of control over and complexity of the technology given to the children involved is appropriate to prevent overwhelm. However, the selected groupings of children in this study were either primarily neurodivergent (i.e. diagnosed with ADHD, autism, Down's syndrome, or learning differences) or were composed of

children who attended special educational provisions due to differences in social and cognitive styles. Due to this, the insights derived from this study focus predominantly on the needs of children with impairments[13]. In our research, we aim to expand upon this by focusing on understanding the design space for mixed groups of autistic and neurotypical children.

2.3 Considering the Double Empathy Problem in Designing for Mixed Groups

Ten years ago, Milton proposed the 'double empathy problem' to explore the interpersonal issues between autistic and neurotypical peers[22]. At its core, the double empathy problem outlines the barriers that might emerge from the social interactions between different groups of people that are not inherent to any one group. Rather, they are based on mutual interpersonal issues[21]. For example, an autistic child's use of body language instead of verbal communication would not be the only barrier to explaining themselves to a neurotypical peer but the peer's lack of understanding or different method of communication through accustomed verbal means also contributes. A recent panel on the double empathy problem with 15 researchers revealed that this concept could assist to bridge the gap between autistic and neurotypical peers and should be utilized for future autism knowledge production [21]. In this regard, we propose to use the double empathy problem as a 'concept to think and design with', while exploring how it could aid our research and support the understanding of mutual interpersonal issues and interactions in inclusive social play for mixed groups. We should note that the double empathy problem does not currently present practical guides or parameters to apply to research findings. However, we argue it is a concept to think with when designing for mixed groups.

3 METHODOLOGY

We conducted semi-structured interviews with 6 experienced professionals who work with both autistic and neurotypical children at mainstream primary schools (P4, P5, P6), city councils (P1), disability support organizations (P2), and gyms which promote inclusive physical activities (P3). Their roles within their respective institutions are Special Educational Needs Coordinators (SENCO) (P1, P4, P6), Assistant Head Teacher (P5), Head of Youth Services (P2), and Manager (P3), and they have 7 to 30 years of experience working with mixed groups ($M=19.33$, $STD=9.95$). Their institutions support 183 to 800 children ($M=528.60$, $STD=239.90$) from ages 4 to 12, of which 10 to 24 are diagnosed with autism or in the process of diagnosis ($M=17.67$, $STD=6.44$). We recruited participants from the Bristol City Council mailing list and through registered primary schools. Then, we proceeded with snowball sampling. We conducted the interviews using Microsoft Teams, which lasted 45 minutes each. Each interview was audio-recorded and transcribed. We asked questions about participants' experience with working with mixed groups, the forms, and interactions of inclusive social play, the challenges both groups face when they engage in social play, and their strategies to address these challenges. We used thematic analysis to analyze the transcripts[10]. The first two authors separately coded each interview via inductive coding. All authors discussed the resulting codes. Then, the first two authors coded

all transcripts together following consensus and created the initial themes, which we refined and finalized again through discussion sessions between all authors following a typical peer validation process[14].

4 RESULTS

We should note that our findings from the interviews presented in this section are the interpretations of the professionals' experiences and not of the children themselves. While the questions we asked participants were open-ended to include barriers for both autistic and non-autistic children (i.e. "What is the engagement or interaction usually like between autistic and non-autistic children in these play sessions?") and all the participants worked in mixed group environments, we found that they tended to focus on the barriers they believed autistic children faced when engaging in mixed group play. Given this tendency and the previous discussion in the paper, we have also attempted to understand other barriers to mixed group social play beyond those specifically identified for autistic children. In this sense, we have attempted to use the 'double empathy problem' as a lens to consider other aspects or challenges of interactions in mixed group social play activities. In the following, we present the themes identified from the interviews, which include the perceived barriers autistic children face in inclusive social play and the solutions currently implemented by the participants. Then, within the subsection and in italics, we present our own understanding in relation to the parallel barriers or issues that neurotypical children in mixed group settings may encounter by incorporating ideas from the double empathy problem (Figure 1). These serve the purpose of understanding the mutual interpersonal issues in inclusive social play in mixed groups, which we later utilized to create and discuss design insights (see Discussion).

4.1 Differences in Communication Styles

Participants reflected on the various ways autistic children communicate and due to the differences in communication methods, they believed it may be difficult for autistic children to convey their needs to others. Participants noted how they believed autistic children's communication may appear through behavior, such as hitting or poking, and is specific to each child. Participants commented that they look out for these behaviors and have learned how to interpret their meanings, to ensure the child is being listened to and understood. Another solution mentioned for these communication differences was to find a common communication ground, often through using signs or visuals. P6 discussed how many children at their school were generally understanding of these kinds of differences and had an awareness that some children may communicate differently. However, two participants expressed that they believed these communication differences may result in autistic children experiencing difficulties when attempting to initiate play. P4 stated that they interpreted one child's behavior of prodding other children as a method of getting attention and asking to play together, which they observed was not always understood by other children. *"For some [autistic] children, it can be something like they hit but they don't hit because they want to be violent, they're hitting because that is their way of communicating."* - P2.

ACTORS		DESIGN SPACE OF INCLUSIVE SOCIAL PLAY FOR MIXED GROUPS			
Autistic Children	Themes	Differences in Communication Styles	Understanding Ambiguous Rules or Norms	Emotional and Sensory Regulation	Strong Desire to Play on Own Terms
Neurotypical Children	Double Empathy Reflection	Following Accustomed Methods of Communication	Diverging from 'Correct' Rules.	Empathizing with Sensory Perception Differences	Expectation to Act as a Cohesive Group
Mixed Groups	Design Insights	Create a Common Medium of Communication and Interaction		Personalize and Modularize Modalities	

Figure 1: The summary of the design space, including the themes, double empathy reflection, and design insights according to the mixed group inclusive social play actors, autistic and neurotypical children.

4.1.1 Following Accustomed Methods of Communication. Neurotypical children may expect others to communicate a certain way or may have preexisting ideas on how a conversation should unfold from previous experiences. These previous experiences may come from primarily neurotypical interactions and may not include individuals who use different forms or modes of communication. Interview participants also described how they had observed instances demonstrating how neurotypical children may have pre-defined ideas of what certain behaviors mean, such as hitting which they may deem negative as this matches their previous experiences with neurotypical individuals, which can cause further misunderstanding in mixed groups. Placing these predefined normalized approaches to communication onto autistic interaction can result in a communication mismatch. "I think from a non-disabled child's point of view, it can get quite frustrating when they're speaking to nonverbal Jimmy and they don't understand why Jimmy won't speak back to them." - P2

4.2 Understanding Ambiguous Rules or Norms

Some participants suggested that the ambiguity surrounding situations with undefined rules, such as social rules of a shared game, could lead to misunderstandings between autistic and neurotypical children. Most participants discussed how they employed strategies to teach social skills by modeling successful play in structured sessions and building up to include other children in the play. Two participants discussed how they observed autistic children 'perform well' in these structured settings but may struggle to generalize to other scenarios, particularly during unstructured play. "Quite a lot of them [autistic children] will want to fit in. So they'll learn the [social] rules ... but the rules only apply to a certain situation, don't they?" - P1. In addition, the difficulty surrounding ambiguous play rules was discussed by participants with one highlighting that autistic children are more likely to police the rules that they believe are correct, rather than the true rules. One method mentioned to tackle these differences of opinion was to collaboratively establish the play rules with the children before play begins, giving each child an equal opportunity for their opinion to be heard.

4.2.1 Diverging from 'Correct' Rules. Neurotypical children also may struggle to understand that others may not follow or have the same idea of social norms and friendship as them, particularly if their experience of social interactions comes from environments where the majority of people are neurotypical. Due to this, they may believe that their understanding of social rules and norms is the 'correct' one due to the majority of the group behaving similarly to them. This can cause misunderstandings as autistic children may not act in the way expected by a neurotypical person and vice versa.

4.3 Emotional and Sensory Regulation

All participants mentioned that sensory processing differences may challenge autistic children to adapt and regulate themselves during social play, where many external stimuli are present from the play environment (i.e. hot weather) or other play partners (i.e. noise). Participants mentioned they had observed autistic children experience sensory overload when they were unable to regulate effectively. For example, P4 observed that autistic children sometimes struggle to interpret their own body signals, one example being overexerting themselves during physical play, and may not know how to prevent overstimulation. "Their [autistic children's] ability to regulate those emotions and to recognize those signals or to know when they're having difficulties ... they need to be able to learn regulation skills." - P4. Most participants said they work closely alongside these children, monitor the child's emotional state, and provide support, which may interfere with group social play. The participants employ what they call a 'sensory diet' that regulates the sensory stimuli autistic children are exposed to. However, most of the participants stated that these strategies should be directed according to the personal interests and needs of the children, which are nuanced and specialized to each child.

4.3.1 Empathizing with Sensory Perception Differences. Some neurotypical children may struggle to understand what sensory overload feels like as they may not have experienced this before and may not understand how certain stimuli they feel neutral about can be uncomfortable for others. Additionally, some of the actions of neurotypical children may contribute to creating these uncomfortable stimuli in an environment and the lack of awareness of this can cause issues during social play. An example of this was discussed by P1 who created a quiet area in the playground specifically for those experiencing sensory overload but it was difficult to keep quiet due to the noises produced by other children.

4.4 Strong Desire to Play on Own Terms

Participants mentioned that they observed autistic children preferred doing things a certain way that may not be shared by others, causing difficulties with social play in mixed groups. "They [autistic children] don't want to sit and wait their turn, they just wanna go off and do what they wanna do." - P3. P1 mentioned they had observed that the topic and setting of successful mixed group play is generally determined by the autistic child, as they believed "[autistic children] find it hard to understand what others may need from a situation". Participants mentioned that arguments can arise when a situation does not go exactly the way an autistic child wants it to go or if they feel like their opinion is not being listened to, with

P6 mentioning that compromising is particularly challenging. One solution proposed by P4 involved one autistic child running her own lunchtime mindfulness coloring group alongside other children who also enjoyed the same activity. Additionally, participants observed that some autistic children often appear to be reluctant to play with peers, with P2 describing some children who become distressed when in close proximity to others. To encourage interaction, participants often model successful play by incorporating activities the child enjoys to maintain interest and gradually inviting others into the play space, ensuring the child is comfortable with the children in their environment.

4.4.1 Expectation to Act as a Cohesive Group. Similarly to the above themes, neurotypical children may have limited experience with autistic individuals and may apply their understanding of neurotypical interaction to situations with mixed groups, resulting in a misunderstanding of needs. Neurotypical children may desire to play socially in a particular way, which they assume everyone else shares as other neurotypicals may also want to play similarly. However, there may be autistic members of the group who might have a different idea of what social play and friendship looks like and involves, which can cause misunderstandings. One example P1 discussed is how often neurotypical children in mixed group scenarios found it difficult to understand why autistic children may not follow the shared rules of a game resulting in arguments.

5 DISCUSSION

5.1 Design Insights for Facilitating Inclusive Social Play for Mixed Groups

In this section, we combine our understanding of relevant research in the field, themes from our interviews, and our understanding of the concept of the 'double empathy problem' to present two design insights for facilitating inclusive social play for mixed groups (Figure 1). The following design insights are either novel or are building upon existing insights in the literature for autistic children, which we believe can be extended to social play in mixed group settings.

5.1.1 Create a Common Medium of Communication and Interaction. The interviews revealed, autistic children's ways of communication can be nuanced and may contain different meanings from more typical contexts (e.g. hitting to engage). Yet, most neurotypical children might expect an accustomed method of communication (i.e. verbal) and might have trouble understanding diverging communication styles. One way to address these differences could be to create a common medium of communication that doesn't rely on the interpretation of behavior or language. P5 suggested that *"a child could use something handheld so that they can indicate to other children that they want to play... something that they could use to make it clear to other children what they want to do rather than have to rely on other people's interpretation of what they think they want to do."* Wilson et. al. have previously examined supporting self-expression for autistic children through different modalities which could be extended to mixed group scenarios[31].

Additionally, participants mentioned that shared social play rules may be misunderstood by autistic children. However, most neurotypical children might not realize their autistic peers struggle

to understand and follow these rules. P6 discussed that *"getting the group together to establish the rules so everyone agrees on the rules [has] been quite fundamental in moving forward"*. However, P6 stated that this collaborative discussion around play rules often has to be facilitated by an adult. Designers could help children establish shared play rules through a common medium of interaction to aid mutual understanding and promote independence.

5.1.2 Personalize and Modularize Modalities. The interviews suggested that autistic children's sensory processing is different from their neurotypical peers' and their emotional states may be sensitive to sensory stimuli. However, social play may include many actions with sensory output (i.e. shouting and touching). Neurotypical children may not realize these actions may discomfort their autistic peers. To address this, participants often used the personalized interests of autistic children to engage them in play (i.e. spinning or touching). P2 stated that they *"plan the [sensory] activities around [the] interests [of the children] because we know when children are engaged in play in an activity that they like to do, they're gonna feel more confident, they're gonna feel more comfortable"*. Thus, one approach could be personalizing the sensory modalities and the devices children use for play according to individual preferences. Frauenberger et. al. have previously explored the advantages of providing flexible configurations in social play technologies, enabling each child to customize parts of the technology during play to suit their own interests, ideas, and needs[13].

In addition, participants stated they believe many autistic children are black-and-white thinkers. They may not easily understand what others need from a situation and may force their own point of view. Yet, during social play, most neurotypical children play on group terms rather than any individual's. They might not understand why play should follow one child's preferences and may expect everybody to do the same thing. One example strategy given by P2 to support collaboration was baking a pizza as a group. The children would create the pizza together, but each slice would be allocated to each child according to their own tastes. Similarly, designers could create modular devices that accommodate each child's preferences individually, while allowing them to contribute to group play[13]. When combined, these devices could reflect collaborative play activities as a group. P5 suggested a collaborative game where *"there will be multiple different scenarios and different paths you could take, so it's not just a linear progression, but they could choose which way they want to go and they could split off into individual teams as well as just go off as one big team."*

5.2 Double Empathy Problem in Mixed Group Research & Future Work

The concept of the double empathy problem helped us to consider the social play experiences of individuals across mixed groups of autistic and neurotypical children and create design insights concerning all those involved. However, the practical operationalization of the double empathy problem is not well specified by previous research and there is little practical information on how to apply it within design methodologies. Thus, we were challenged to reflect on mixed group perspectives and should emphasize these reflections are our assumptions based on our findings and experience as neurotypical researchers. Yet, this was an initial application

where we explored and speculated upon the potential of the double empathy problem to shape our research. We believe that we need to develop a framework to both understand the double empathy problem to its full extent and guide researchers like ourselves on how it could be applied. We plan to work with this concept in the next steps of our research, by which we hope to create a methodological approach to apply the double empathy problem and discuss its underlying constructs in mixed group research.

The work presented in this paper is a part of the initial contextual inquiry toward a research project for designing technologies to support and facilitate inclusive social play for mixed groups. In the future, we plan to conduct observation sessions of inclusive social play to reflect on our learning from this study, conduct participatory design workshops to apply the presented design insights to play technologies together with autistic and neurotypical children, and evaluate these technologies to explore how to design for inclusive social play for mixed groups.

6 CONCLUSION

We aimed to understand the design space to support and facilitate inclusive social play for mixed groups, by conducting semi-structured interviews with 6 professionals who work with both autistic and neurotypical children. Despite our focus on mixed groups in the interviews, our participants still tended to focus on barriers to inclusive social play from the perspectives of autistic children. We proposed and utilized the concept of the double empathy problem to address this and consider the neurotypical children's perspective in inclusive social play. Accordingly, we presented themes and design insights that highlight challenges to inclusive social play for mixed groups from both perspectives of autistic and neurotypical children. We argue such an approach raises important questions about design methods in this space.

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